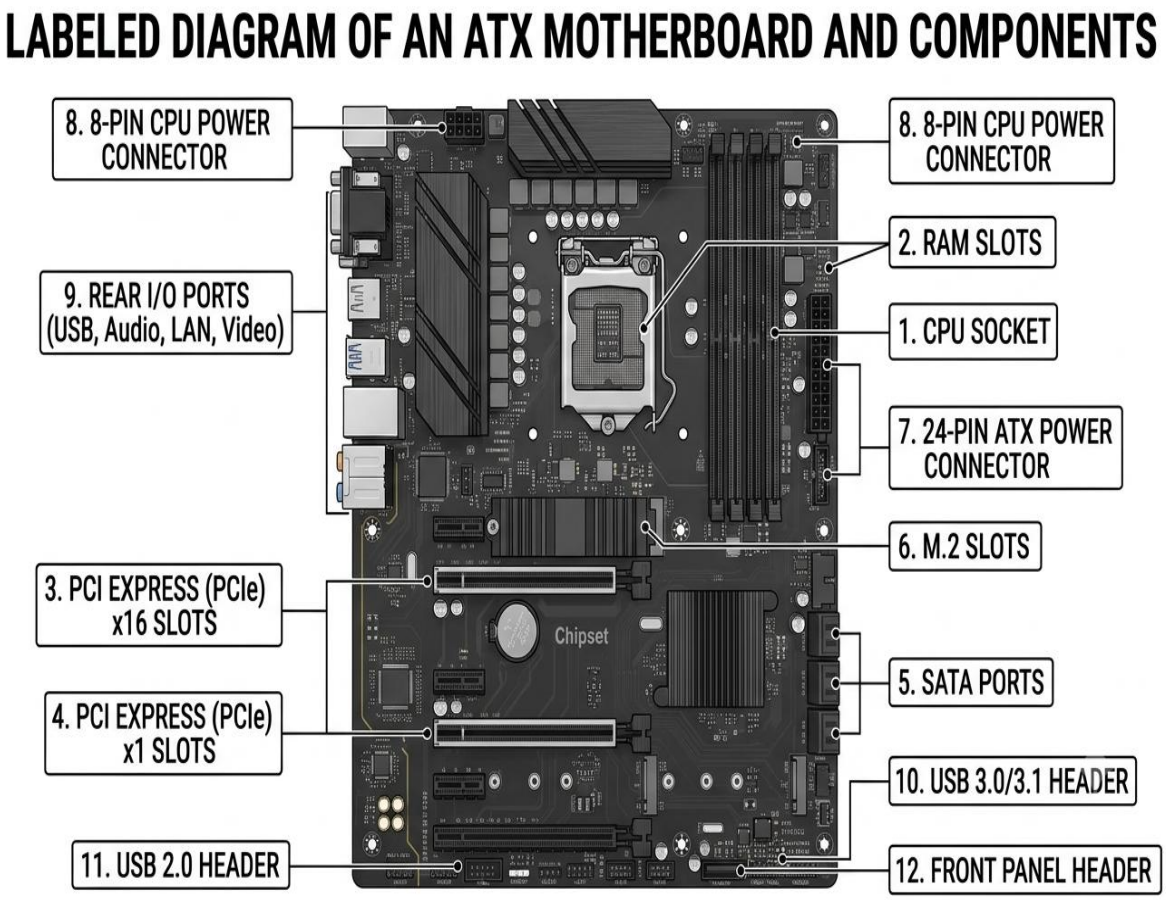
1) **Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.)**



2) **Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.**

Here are three popular CPU socket types along with a compatible CPU model for each:

1. AMD Socket AM4

- **Compatible CPU Model:** AMD Ryzen 7 5800X3D (or Ryzen 5 5600X)
- **Details:** AM4 is one of AMD's most successful and long-lasting platforms, utilizing a Pin Grid Array (PGA) design where the pins are on the processor. It supports DDR4 memory and spans multiple generations of Ryzen processors (Ryzen 1000 through 5000 series).

2. Intel Socket LGA 1151

Details : LGA 1151 is a desktop CPU socket introduced by Intel in 2015 to replace LGA 1150. It was the standard socket for four generations of Intel Core processors:

- **Generations Supported: 6th (Skylake), 7th (Kaby Lake), 8th (Coffee Lake), and 9th Gen (Coffee Lake Refresh).**

Compatible CPU Models

Depending on the socket version and motherboard chipset, compatible CPU models include:

1. Flagship / High-End (9th Gen - LGA1151 v2 / 300-series boards):
    - Intel Core i9-9900K
    - Intel Core i7-9700K
2. Popular Mid-Range (8th Gen - LGA1151 v2 / 300-series boards):
    - Intel Core i7-8700K
    - Intel Core i5-8400
3. Older Generations (6th & 7th Gen - LGA1151 v1 / 100 & 200-series boards):
    - Intel Core i7-7700K (7th Gen Kaby Lake)
    - Intel Core i7-6700K (6th Gen Skylake)

3. AMD Socket AM5

- **Compatible CPU Model:** AMD Ryzen 5 7600X (or Ryzen 7 7800X3D / Ryzen 9 9950X)
- **Details:** AM5 is AMD's modern desktop platform. Moving away from traditional PGA sockets, AM5 uses an LGA 1718 design where the pins are located on the motherboard socket instead of the CPU. It exclusively supports DDR5 memory and PCIe 5.0.

3) Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

Think of the CPU and the chipset as the ultimate collaborative duo on your motherboard: the CPU is the master intellect doing the heavy thinking, while the chipset is the master logistics coordinator keeping everything connected and talking to each other.

1. The CPU (The Brain)

The Central Processing Unit is the primary computational engine of your computer.

- Primary Functions: It executes instructions from software and operating systems, performs complex mathematical calculations, makes logical decisions, and directs the flow of data across the system. Everything from rendering graphics to running background code relies on the CPU's raw processing power.

2. The Chipset (The Traffic Controller)

The Chipset (often called the Platform Controller Hub, or PCH, in modern systems) is a collection of chips on the motherboard that acts as a communication hub.

- Primary Functions: While modern CPUs have taken over some tasks that used to belong to the chipset (like managing the RAM and primary graphics card), the chipset still manages the vast majority of input/output (I/O) traffic. It controls communication between the CPU

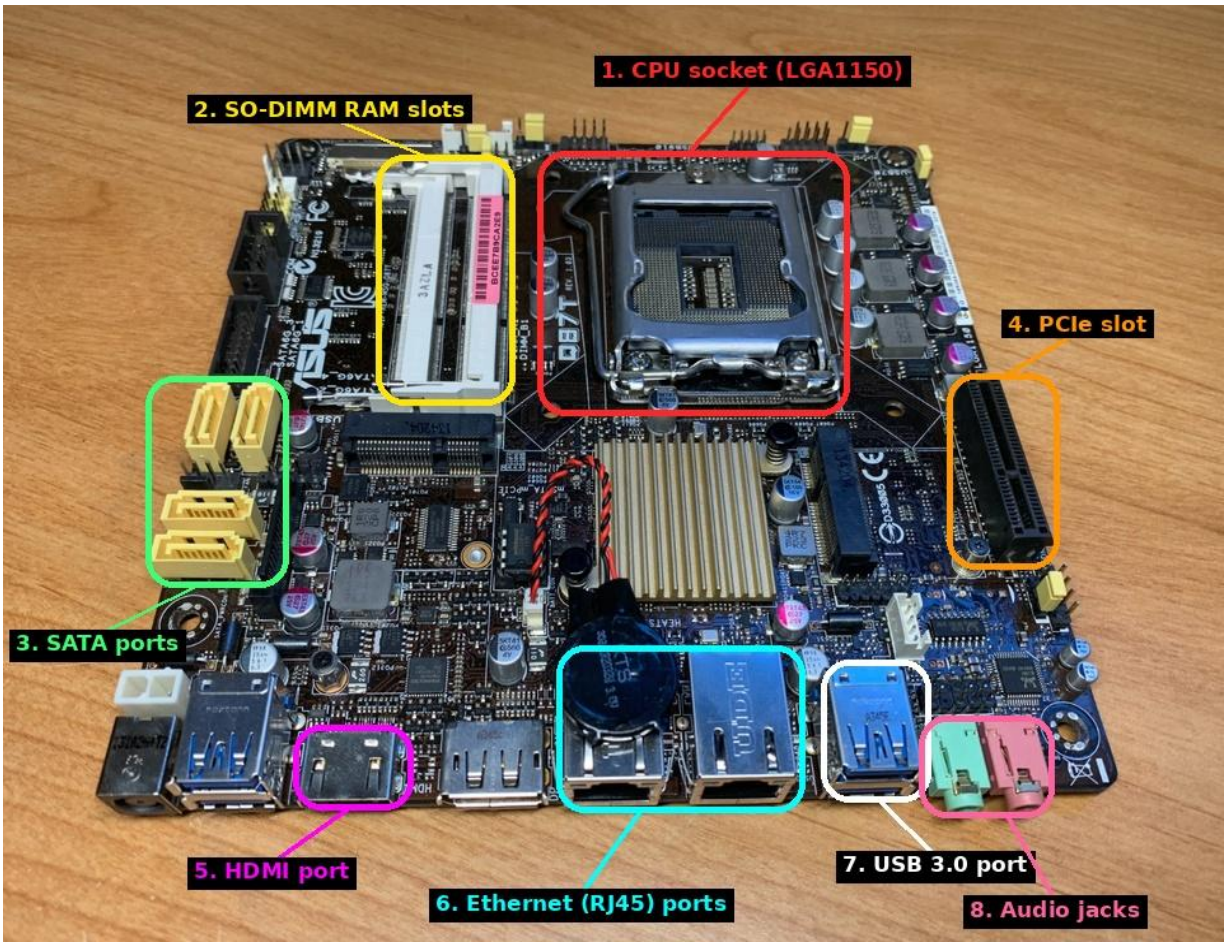
and peripheral devices, such as USB ports, SATA storage drives, audio controllers, network adapters, and secondary PCIe slots.

How They Work Together: Opening Spotify

When you click on the Spotify app icon, the CPU and chipset engage in a synchronized dance to bring your music to life:

1. **The Click (Input & Routing**): When you click your mouse, the click signal travels through a USB port. The chipset receives this signal, processes it, and routes it up to the CPU, letting it know that you want to open Spotify.
2. **Retrieving the App Data (Storage & Memory**): The CPU tells your storage drive (an SSD) to fetch the Spotify program files. While high-speed NVMe drives often talk directly to the CPU, the chipset coordinates other data traffic, ensuring the compressed app files are read from storage and loaded into your system's temporary memory (RAM).
3. **Processing and Launching (Execution**): Once the files are loaded into RAM, the CPU begins fetching and executing those instructions. It decodes the application code, allocates processing threads to run the app, and calculates how the user interface should look, working alongside your graphics card (GPU) to draw the Spotify window on your screen.
4. **Streaming and Audio Output**: Once Spotify is open and you hit play on a song, the CPU decodes the compressed audio stream (like Ogg Vorbis or MP3 data coming over your internet connection, which the chipset routed through your Wi-Fi/Ethernet card). Finally, the CPU sends the decoded audio data to the audio controller or a USB/Bluetooth audio device managed by the chipset, delivering sound smoothly to your headphones or speakers.

4) **Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.**



1. **CPU socket (LGA1150):** the large socket at the top center with the metal lever and retention frame. The “Q87T” model name is printed next to it.
2. **SO-DIMM RAM slots:** the white slots at the upper left. They take the shorter laptop-style memory sticks, which suit this compact board.

3. **SATA ports:** the four yellow connectors on the left edge, for hard drives and SSDs.
4. **PCIe slot:** the black slot on the right edge, for an expansion card such as a graphics card.
5. **HDMI port:** on the rear I/O panel at the bottom left, for video output to a monitor or TV.
6. **Ethernet (RJ45) ports:** the two network jacks at the bottom center. This board has two.
7. **USB 3.0 port:** the blue port at the bottom right. Blue usually means USB 3.0.
8. **Audio jacks:** the green and pink 3.5 mm jacks on the right of the rear panel. Green is for speakers or headphones and pink is for a microphone.